

Phishing Website Detection Using Random Forest and NLP-Based URL Analysis

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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1. Abstract:

Phishing websites are a major cybersecurity threat that trick users into revealing sensitive information such as usernames, passwords, banking credentials, and personal data. As phishing attacks become more advanced, traditional blacklist-based detection methods are often unable to identify newly created malicious websites. This project proposes a machine learning-based phishing detection system using the Random Forest algorithm combined with Natural Language Processing (NLP)-based URL analysis to accurately classify websites as legitimate or phishing. The proposed system uses a dataset containing both legitimate and phishing URLs. During preprocessing, URLs are cleaned and analyzed using NLP techniques such as tokenization and lexical feature extraction. Features including URL length, number of subdomains, number of special characters, suspicious keywords, number of digits, and HTTPS usage are extracted. The textual data is then converted into numerical form using Term Frequency-Inverse Document Frequency (TF-IDF), making it suitable for machine learning. The extracted features are used to train a Random Forest classifier, which is chosen for its high accuracy, robustness, ability to handle large feature sets, and reduced risk of overfitting. The model is evaluated using performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix to measure its effectiveness. The proposed system provides a fast, scalable, and reliable solution for phishing website detection. It can be integrated into web browsers, email filtering systems, and cybersecurity applications to improve online security and protect users from evolving phishing attacks.

2. INTRODUCTION:

Phishing is a common cyberattack in which attackers create fake websites that imitate legitimate ones to steal sensitive information such as login credentials, banking details, and personal data. As phishing techniques continue to evolve, traditional blacklist-based detection methods often fail to identify newly created malicious websites. Machine Learning (ML) offers an effective solution by detecting phishing websites based on learned patterns rather than manually maintained rules. Among various ML algorithms, Random Forest is known for its high accuracy, reliability, and ability to handle complex datasets. Additionally, Natural Language Processing (NLP) helps analyze URL text by extracting lexical patterns and suspicious keywords that may indicate phishing attempts.

This project aims to develop a phishing website detection system using Random Forest and NLP-based URL analysis. The system extracts textual and lexical features from URLs, converts them into numerical representations using Term Frequency-Inverse Document Frequency (TF-IDF), and classifies URLs as either legitimate or phishing. The proposed solution provides an **accurate**, scalable, and efficient approach to detecting phishing websites, helping improve online security and protect users from cyber threats.

3. SOFTWARE REQUIREMENTS:

CATEGORY	SPECIFICATION
Operating System	Windows 11 (64-bit
Processor	Intel Core i5
RAM	Minimum 8 GB
Storage	20 GB Free Disk space
Programming Language	Python 3.11 or above
Development Environment	Jupyter NoteBook
Machine Learning Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib
Dataset	Phishing and Legitimate URL Dataset (Kaggle)
Machine Learning Algorithm	Random Forest, Logistic Regression, Support Vector Machine (SVM), XGBoost
Backend Framework	Flask
Frontend Technologies	HTML, CSS, JavaScript
Version Control	Git and GitHub

Signature of the Guide

Course Instructor